

Slide 1: Welcome to jQuery

"Write Less, Do More"

What is jQuery?

jQuery is a fast, small, and feature-rich JavaScript library. It simplifies things like HTML document traversal (finding elements), event handling, and animations. Tasks that take 10 lines of code in Vanilla JavaScript can often be done in just 2 lines with jQuery.

Why are we learning it?

- **Simplicity:** It uses CSS-like selectors, making it very intuitive.
 - **Compatibility:** It handles cross-browser issues (like IE) automatically.
 - **Market Share:** Approximately 70% of the world's top websites still utilize jQuery in their stack.
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Slide 2: Installation & Ready Event

Step 1: The CDN

Add this script tag at the end of your `<body>` section to load the library:

HTML

```
<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
```

Step 2: The Document Ready

Explanation: We want to ensure our JavaScript runs only after the full HTML document has loaded. Otherwise, jQuery might try to find an element that doesn't exist yet.

Code Example:

JavaScript

```
$(document).ready(function() {  
    console.log("The DOM is fully loaded and ready!");  
});
```

// Short-hand version:

```
$(function() {  
    console.log("Ready to go!");  
});
```

Slide 3: Basic Selectors (The Heart of jQuery)

Concept:

The primary function of jQuery is to "select" an element using the `$()` syntax.

- `$("p")` → Targets all `<p>` (paragraph) tags.
- `$("#myId")` → Targets a unique element by its ID.
- `$(".myClass")` → Targets all elements with a specific class.

Task List (Basic):

- **Task 1:** Change the text color of an `<h1>` to "blue" using a selector.

- **Task 2:** Hide all `<li>` items when a button is clicked.
- **Task 3:** Set the font-weight to "bold" for elements with the class `.highlight`.

Solution for Task 1:

JavaScript

```
$("#h1").css("color", "blue");
```

Slide 4: Text, HTML & Value Functions

Explanation:

- `.text()`: Sets or returns the plain text content of selected elements.
- `.html()`: Sets or returns the content of selected elements, including HTML tags.
- `.val()`: Used to get or set the value of form input fields.

Task List (Medium):

- **Task 1:** Change the text of a heading to "Updated!" upon clicking.
- **Task 2:** Insert a `<b>Hello World</b>` tag into an empty div.
- **Task 3:** Capture the text from an input field and display it in an alert.
- **Task 4:** Clear an input field's value when a "Reset" button is clicked.

Solution for Task 3:

JavaScript

```
$("#btn").click(function(){  
    let myVal = $("#input").val();  
    alert(myVal);  
});
```

Slide 5: Attributes & CSS Manipulation

Explanation:

- `.attr()`: Used to change attributes like an image's `src` or a link's `href`.
- `.css()`: Used to apply inline styling directly to elements.

Task List (Interactive):

- **Task 1:** Change an image `src` when a button is clicked (Switch Image).
- **Task 2:** Change both the `background-color` and `border-radius` of a box simultaneously.
- **Task 3:** Get the `href` of a link and print it to the console.
- **Task 4:** Increase a heading's font-size to "50px" when it is clicked.

Solution for Task 2:

JavaScript

```
$("#box").css({  
  "background-color": "orange",  
  "border-radius": "10px"  
});
```

Slide 6: Effects & Chaining

Explanation:

jQuery allows you to run multiple actions on the same element in a single line of code.

This is called **Chaining**.

- **Core Effects:** `.hide()`, `.show()`, `.toggle()`, `.fadeOut()`, `.fadeIn()`

Task List (Visual):

- **Task 1:** Make a box `fadeOut` and then immediately `fadeIn`.
- **Task 2 (Chaining):** Hide a heading, change its color to red, and then show it again all in one line.
- **Task 3:** Create a "Toggle" button that shows or hides a paragraph.
- **Task 4:** Display an alert only after a `.hide()` animation has completely finished.

Solution for Task 2:

JavaScript

```
$("#h1").hide(1000).css("color", "red").show(1000);
```

Final Practical Activity

Project: "The Smart Profile Card"

Build a card containing a User Name and an Image. Create 3 buttons below it:

1. **"Dark Mode":** Toggle the card's background to black and text to white.
2. **"Change User":** Update the Name text and the Image source.
3. **"Surprise":** Use an animation to make the card disappear and reappear.